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PALAEOMAGNETISM OF THE LOWER CRETACEOUS VULCANITAS CERRO COLORADO FORMATION OF THE SIERRA DE LOS CÓNDORES GROUP PROVINCE OF CÓRDOBA, ARGENTINA

Daniel Alberto VALENCIO

Departamento de Ciencias Geológicas, Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires, Ciudad Universitaria, Nuñez, Buenos Aires, Argentina

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Measurements are described of the directions of magnetization of 101 samples from six lava flows, one dyke and redbeds from the Lower Cretaceous Sierra de los Cóndores Group, Argentina.

The magnetic remanence of the lowest basaltic flow (Embalse Río Tercero Formation) was unstable. Primary remanent magnetizations of the other five lava flows, the dyke and redbeds (Vulcanitas Cerro Colorado Formation) were isolated after detailed A.C. and thermal cleaning; all these units have reversed magnetization. On the basis of the mean direction, referred to the bedding, of cleaned remanent magnetization of each unit a virtual geomagnetic pole is computed; the mean of these virtual geomagnetic poles, SAK_2, is: 14°E 81° S, $\alpha_{95} = 13°$; we think that SAK_2 is a geomagnetic pole.

The position of the pole for the Vulcanitas Cerro Colorado Formation (118.5 ± 6 my) is reasonably close to the Serra Geral Formation pole (115–125 my); this supports the interpretation that South America and Africa were separated in Lower Cretaceous time and suggests that the earth magnetic field was, on average, similar to that of a geocentric dipole in the Lower Cretaceous.

The reversed remanent magnetization and the K–Ar age (118.5 ± 6 my) of the Vulcanitas Cerro Colorado Formation define the existence of a previously unrecognized period of reversed polarity at the Lower Cretaceous.

1. Introduction

Late Palaeozoic and Early Mesozoic geomagnetic poles for South America and Africa overlap when these continental blocks are joined by their Atlantic littorals [1]. In this palaeogeographic reconstruction the South American geomagnetic poles for Chon Aike [2] and Serra Geral Formations [3] do not superpose with their contemporaneous African geomagnetic poles, suggesting either a pre-Middle Jurassic (161 my) or a pre-Middle Cretaceous (113–125 my) age respectively, for the formation of the South Atlantic Ocean. However, each one of these two possible interpretations is based on only one South American geomagnetic pole. More Jurassic and Cretaceous geomagnetic poles for South America are necessary to see which of these two interpretations is the more reliable.

Palaeomagnetic studies of South American Cretaceous rocks have been recently reported [4]. However, neither A.C. nor thermal cleaning have been carried out and the results have been accepted only provisionally; this is the case of the palaeomagnetic study of dykes at Herradura near Lima, Perú; the Aptraxa Sandstone, Colombia and the Yeguera and Cazuela Formations, Venezuela [4]. Therefore the palaeomagnetic poles computed on the basis of these studies are not adequate for palaeogeographic interpretations. More recently a virtual geomagnetic pole for the Cretaceous La Teta lava and Maruayan diorite from the Guajira Peninsula of Colombia, has been reported by MacDonald and Opdyke [5]; this pole is about 90 degrees from the Serra Geral pole. On the basis of this result the authors have suggested a tectonic rotation for the Guajira area.

Lower Cretaceous igneous and sedimentary rocks crop out at the Sierra de los Cóndores and Río Tercero, in the southern part of Córdoba Province, Argentina. Gordillo and Lencinas [6], have included rocks of the first mentioned area within the Sierra de los Cóndores Group. Particularly, the volcanic rocks

of this Group are considered as having originated in an igneous cycle contemporaneous with the one in which Serra Geral volcanics were formed. In this paper results of the palaeomagnetic study of the igneous and sedimentary rocks of the Vulcanitas Cerro Colorado Formation of the Sierra de los Cóndores Group, are given. Palaeomagnetism of the overlying Vulcanitas Rumipalla Formation of the same Group and the igneous rocks from Río Tercero will be given in separate papers. The contribution of these three studies to the stratigraphic correlation of these rocks will be discussed in another paper.

2. Available data

2.1. *Geological evidences*

The Sierra de los Cóndores belongs to the Sierras Pampeanas System; it is situated in the south of Córdoba Province, Argentina. In this area a sequence of redbeds and volcanic rocks, about 200 m thick, of Late Lower Cretaceous age, rests with angular unconformity on the plutonic-metamorphic basement [6]. These rocks have been classified in four formations named in chronological order: Embalse Río Tercero Formation, Vulcanitas Cerro Colorado Formation, Conglomerado Cerro Libertad Formation and Vulcanitas Rumipalla Formation; all these formations form the Sierra de los Cóndores Group [6].

The sedimentary episode during which the Embalse Río Tercero Formation was formed, was interrupted by a volcanic cycle. During this cycle more than seven lava flows were formed; sediments which were laid down between the lava flows have the same characteristics as those formed before the volcanic cycle [6]. The Vulcanitas Cerro Colorado Formation include the lava flows extruded in this first volcanic cycle and their interbedded continental sediments. Rocks of this formation are fairly well exposed in the Cerro Colorado area, about 32° South, 64° West.

The petrographic characteristics of the volcanic rocks of the Sierra de los Cóndores Group are described by Gordillo and Lencinas [6]. According to these authors six lava flows and one diabasic dyke are exposed at the Cerro Colorado area. Mentioned in chronological order, and according to the petrographic study, the lava flows are: (1) Picrite basalt; (2) Calc-alkalic trachyte, lower flow; (3) Black-trachybasalt lower flow; (4) Calc-alkalic trachyt, upper flow; (5) Brown-trachybasalt, and (6) Black-trachybasalt, upper flow, fig. 1.

The Picrite basalt belongs to the Embalse Río Tercero Formation; the other five lava flows and the dyke belong to the overlying Vulcanitas Cerro Colorado Formation. The authors mentioned above suggest that the lowermost lava flow of this Formation exposed at Cerro Colorado (Calc-alkalic trachyte, lower flow) is not the first one of the volcanic cycle, fig. 1.

Samples of the Black-trachybasalt, lower flow, were dated by the K/Ar method (whole-rock) at two different laboratories of the USSR, giving ages of 117 my (Khlopin Radium Institute) and 120 my (Laboratory of Mathematic Geology), respectively [6]. In this paper the possible errors in the determined ages are not quoted. In a good determination, the instrumental and analytical errors are considered to be of the order of 3%. Agreement between the ages of independent samples obtained in two different laboratories can be taken as strong evidence supporting the validity of the potassium-argon age of the Black-trachybasalt, lower flow. Effects such as 'excess' argon and argon loss are more likely to produce discordant ages from samples of the same geological unit. Therefore, a reasonable estimate of the accuracy of the dating of the samples from the Black-trachybasalt, lower flow, is 5%. The mean potassium-argon age for this flow of the Vulcanitas Cerro Colorado Formation is therefore 118.5 ± 6 my. This mean value would be supported by the K/Ar age (whole-rock, 116 ± 6 my) of the trachybasalt from the Cerro Quebracho, which is contemporaneous with that from the Cerro Colorado (Dr. Gordillo, private communication). Recently, K/Ar ages for igneous rocks from Province of Córdoba, which were formed during the same volcanic cycle as the Sierra de los Cóndores Group, have been given [7]. The mean potassium-argon age for this volcanic cycle is 121 ± 4 my, including nine radiometric ages from 112 to 129 my.

2.2. *The palaeomagnetic study*

Sampling was carried out on the basis of the geological study mentioned in subsect. 2.1; more than 100 oriented hand samples of igneous and sedimentary rocks from the Vulcanitas Cerro Colorado and the Embalse Río Tercero Formations were collected

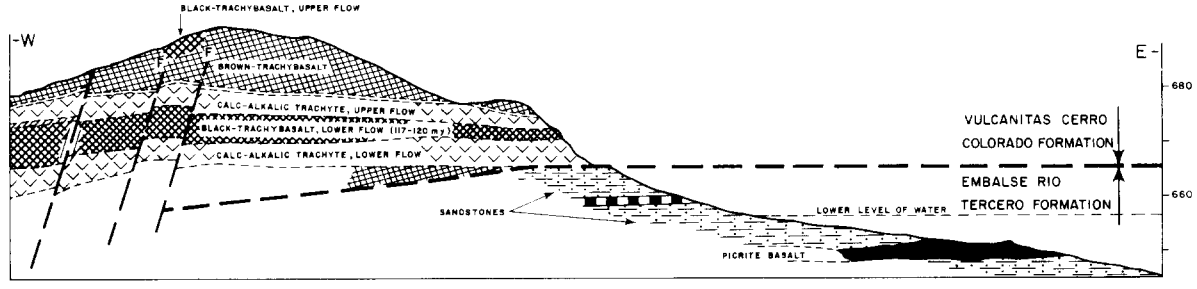


Fig. 1. Schematic geological section at Cerro Colorado, according to Gordillo and Lencinas [6]. The Embalse Rio Tercero and Vulcanitas Cerro Colorado Formations are part of the Sierra de los Cóndores Group.

at the Cerro Colorado area. The orientation of samples was carried out using both the sun and a magnetic compass.

The low level of water in the artificial lake around the Cerro Colorado allowed us to collect samples of the Picrite basalt; normally exposures of this basalt are covered by water, fig. 1. However the rock is badly weathered and, as will be seen, its natural remanent magnetization is unstable.

Sediments baked by lava flows allowed tests of the stability of the natural remanent magnetization. Bedding planes were measured in sediments and in contact between volcanic and sedimentary rocks.

Four disks 2.5 cm in diameter and 1 cm thick were cut of each hand sample. Natural remanent magnetization of these disks was measured using spinner and flux-gate magnetometers. The directions of natural remanent magnetization of samples from each geological unit were highly scattered, but after adequate cleaning the remanent magnetization of samples from most of the units fell into compact groups. Detailed AC and thermal cleaning were used to determine the stability of the remanent magnetization. One disk of each sample of igneous rock and baked sediment was subjected to AC demagnetizing fields increasing in steps of 50 Oe to 300 Oe at least; in this way the best AC demagnetizing field isolating the thermoremanent magnetization of each sample was determined. The criterion adopted for the choice of this best AC demagnetizing field strength was that the direction of remanent magnetization of the pilot disk exhibited no further change in response to treatments in higher demagnetizing field strengths. The best AC demagnetizing field strengths varied from 50 to 250 Oe, depending on the sample. A second and sometimes a third disk of the same sample was submitted to this AC demagnetizing field for a better definition of the direction of the thermoremanent magnetization of the sample. The highly scattered directions of natural remanent magnetization were often found to be associated with high remanent intensities which were sharply reduced after treatment in AC fields of about 100–150 Oe. This was particularly valid for most of the samples from the Calc-alkalic trachyte, upper and lower flows. These samples after the treatment with

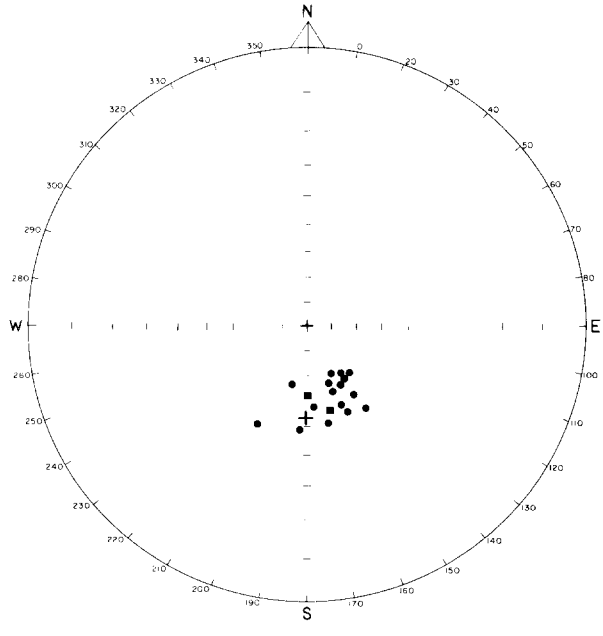


Fig. 2. Directions of cleaned remanent magnetization of hand samples of the Brown-trachybasalt (circles) and of the underlain baked sediments (squares). Solid symbols indicate downward dipping magnetizations (positive inclinations).

AC demagnetizing fields of 300 Oe have remanent magnetization intensities less than 0.1 of their natural remanent magnetization intensities, ($J_r/J_{r_0} < 0.1$). High AC coercivities were found to be associated with samples with stable natural remanent magnetization. This was particularly the case for most of the samples from the Brown-trachybasalt and the Black-trachybasalt, lower flow. These samples keep more than 0.2 of their natural remanent magnetization after the treatment in an AC field of 300 Oe ($J_r/J_{r_0} > 0.2$). The stability of the remanent magnetization of redbeds was verified by a similar detailed thermal cleaning. A weak secondary remanent magnetization was present in redbeds, but it was easily destroyed after a treatment to 100°C. Remanent intensities of redbeds exhibited a sharp reduction between 500°C and 580°C. The statistical method of Fisher [8] was used to compute the mean direction of the primary remanent magnetization of each rock unit, giving unit weight to the direction of the cleaned n.r.m. of each one of its samples.

Samples of Picrite basalt (Embalse Río Tercero Formation) did not give reliable results; they have unstable natural remanent magnetization. However they show predominantly upward inclination. The other five lava flows, the dyke and redbeds (Vulcanitas Cerro Colorado Formation) show stable primary remanent magnetization. Particularly the stability of the remanence of the Brown-trachybasalt was supported by the underlain baked sediments: they show remanent magnetization of the same direction and polarity as the basalt, fig. 2.

Fig. 3 shows the directions of cleaned remanent magnetization of samples of the following units: Black-trachybasalt, upper flow (a); Redbeds (b); Calc-alkalic trachyte, upper flow (c); Black-trachybasalt, lower flow (d); Calc-alkalic trachyte, lower flow (e) and diabasic dyke (f). In table 1 the mean directions of the cleaned remanent magnetization of each rock unit after bedding plane correction are given. However the fold test cannot be applied here because all the samples come from one area where the dip of the strata is uniform. The five lava flows, the dyke and redbeds of the Vulcanitas Cerro Colorado Formation, have all reversed cleaned remanent magnetization. Virtual geomagnetic poles for each one of these rock units are also given in table 1. Fig. 4 shows these virtual geomagnetic poles (triangles) together with South American Triassic and Permo-Triassic ($SATr_{1-5}$), Jurassic (SAJ_1, (2) and Cretaceous (SAK_1, (3) geo-

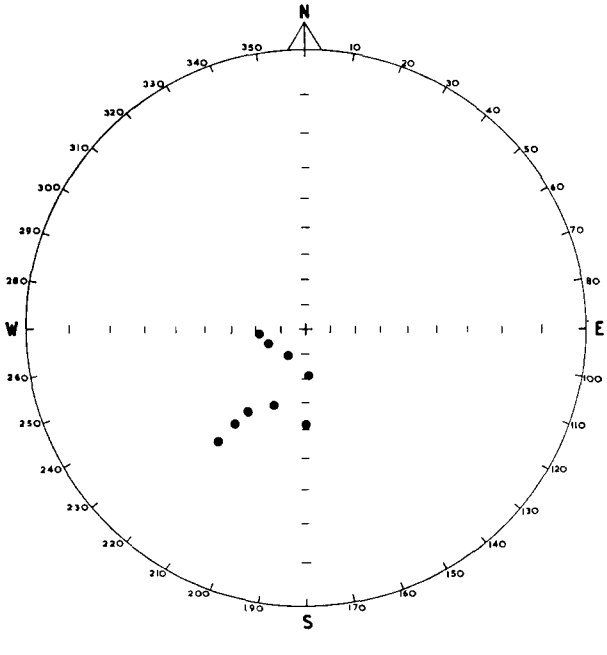


Fig. 3a.

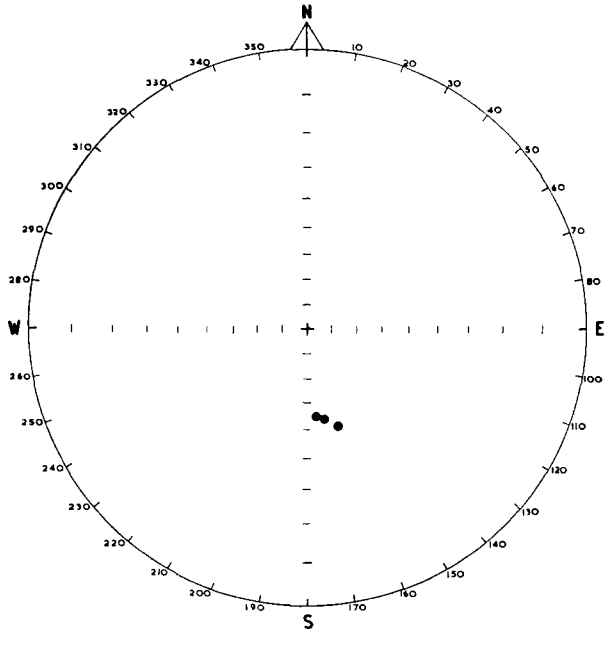


Fig. 3b.

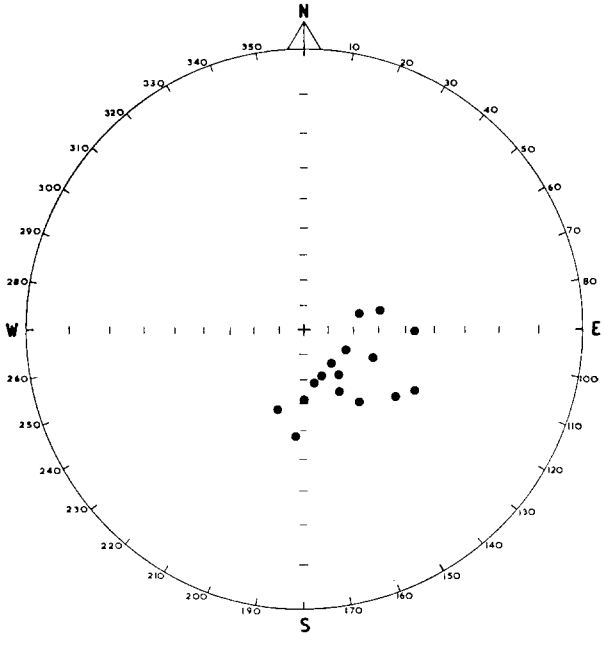


Fig. 3c.

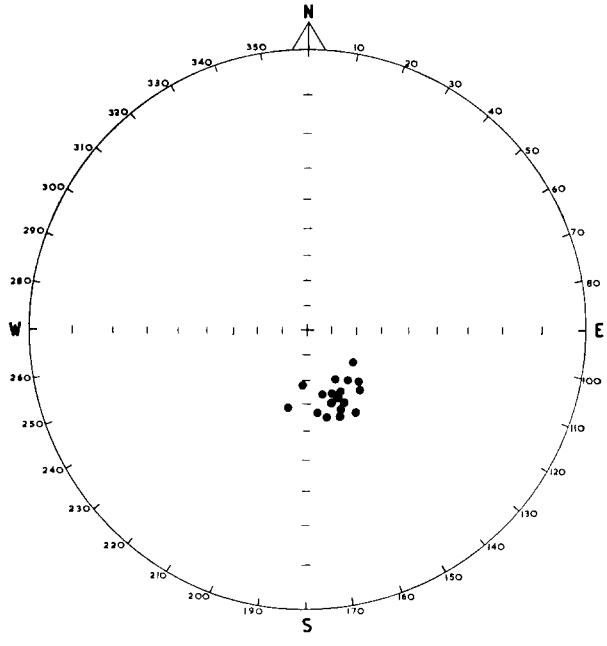


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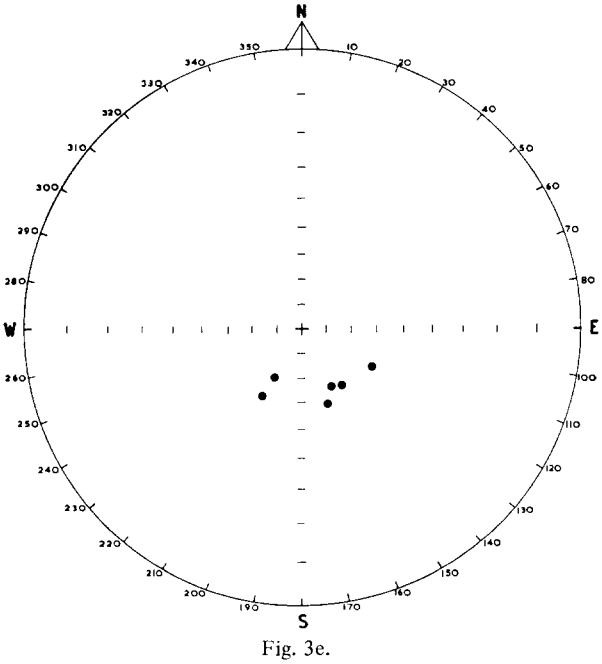


Fig. 3e.

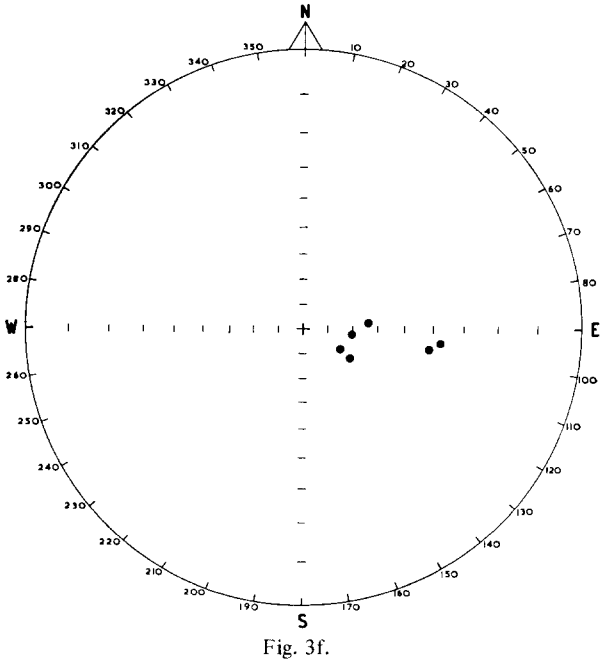


Fig. 3f.

Fig. 3. Directions of cleaned remanent magnetization of hand samples of the: (a) Black-trachybasalt, upper flow; (b) Redbeds; (c) Calc-alkalic trachyte, upper flow; (d) Black trachybasalt lower flow; (e) Calc-alkalic trachyte lower flow and (f) diabasic dyke, from the Vulcanitas Cerro Colorado Formation, Córdoba Province, Argentina.

Table 1
Summary of the palaeomagnetic data of the Vulcanitas Cerro Colorado Formation of the Sierra de los Condores Group, Province of Cordoba, Argentina.

Group	Formation	Rock unit	Number of samples used	Directions of cleaned n.r.m. referred to the palaeo-horizontal					Virtual geomagnetic poles				
				N	D(°)	I(°)	α_{95}(°)	Polarity	Name	Lat (°) (South)	Long (°) (East)	dψ (°)	dχ (°)
Sierra de los Condores	Vulcanitas Cerro Colorado	Black-trachybasalt, upper flow	14	9	204	51	12	R	1	69.7	213	10.9	16.2
		Brown-trachybasalt	20	17	164.5	51	4	R	2	76.8	23	3.6	5.4
		Redbeds	3	3	170	42	6	R	3	78.5	63.5	4.5	8.2
		Calc-alkalic trachyte, upper flow	22	16	147	55	9	R	4	63	9	9	12.7
		Black-trachybasalt, lower flow	18	18	162	51	4	R	5	75.6	12.5	3.6	5.4
		Calc-alkalic trachyte, lower flow	9	6	169	55	14	R	6	80	5	14.0	19.7
		Dyke	6	6	117	55	14	R	7	38.7	2	14	19.7
	Embalse Rio Tercero	Picrite basalt	9	-				–					

magnetic poles. The mean of the Triassic and Permo-Triassic poles (SATr, (1) is also shown. The solid curve in this figure represents the fairly well-known Late Palaeozoic section of the South American polar wandering curve (1). Virtual geomagnetic poles, 2, 3, 4, 5 and 6, are particularly close to the Jurassic-Cretaceous polar zone defined by geomagnetic poles SAJ_1 and SAK_1. The virtual geomagnetic pole for the dyke (7), is far away from this zone; Fig. 3f shows that the directions of the cleaned remanence of the samples of the dyke are not similar to those of the other rock units of the Vulcanitas Cerro Colorado Formation; it could be that the intrusion of the dyke took place during a period of unusual deviation of the geomagnetic field from that of the axial dipole, like that present at a reversal of polarity.

The time embraced by rock units 1 to 6, could be of the order necessary to average out the non-dipole components of the geomagnetic field; therefore the mean of the virtual geomagnetic poles 1 to 6 (SAK_2, 14°E 81°S, α_{95} = 13°) could be considered a geomagnetic pole. Fig. 4 shows that the circle of confidence of SAK_2 overlaps those of the Serra Geral (SAK_1) and Chon Aike (SAJ_1) poles; then all these poles could apparently form a group. Particularly, the Vulcanitas Cerro Colorado Formation pole position (SAK_2) is reasonably close to the contemporaneous Serra Geral Formation pole (SAK_1), and the author thinks that both poles define with more precision the Lower Cretaceous pole position relative to South America. However, more Lower Cretaceous South American geomagnetic poles are necessary to define with adequate statistical precision this pole position. Combination of results from the Mlanje Massif (122 my) and Lupata Alkaline Volcanics (109 my) yields an estimate of the palaeomagnetic pole for the Lower Cretaceous of Africa [9]. This pole position differs from those obtained from the Vulcanitas Cerro Colorado Formation and Serra Geral Formation when rotated with South America to Africa so as to reconstitute Gondwana. This supports the interpretation that the South Atlantic Ocean was open in Lower Cretaceous time. Fig. 5 shows the Late

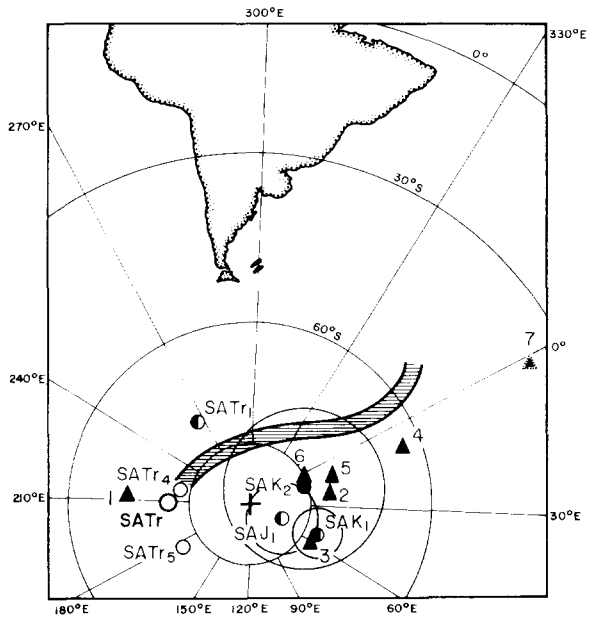


Fig. 4. Mesozoic geomagnetic poles for South America (circles) and virtual geomagnetic poles of the Vulcanitas Cerro Colorado Formation (triangles, Cretaceous 118.5 ± 6 my). SATr, SAJ_1 and SAK_1 are the mean of the Triassic and Permo-Triassic (1), the Jurassic (2) and the Cretaceous (3) poles for South America, respectively. The solid curve is the Late Palaeozoic Section of the polar wandering curve of South America (1).

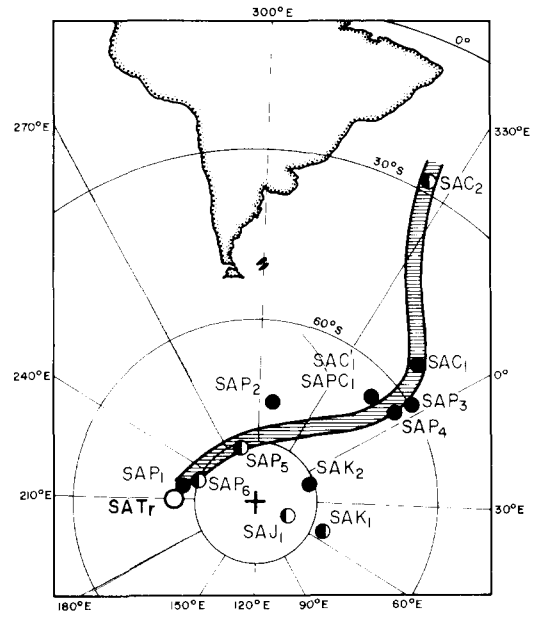


Fig. 5. Late-Palaeozoic and Mesozoic geomagnetic poles for South America according to Valencio and Vilas [1]. SAK_2 is the geomagnetic pole of the Vulcanitas Cerro Colorado Formation. Symbols are shaded according to the polarity of the magnetic remanence of the rocks: white circles indicate normal polarity and black circles reversed polarity. Cretaceous (SAK_1, SAK_2) and Jurassic (SAJ_1) poles apparently define a group independent of the Permo Carboniferous 'age group' (SAC_1, $SAPC_1$, SAP_3, SAP_4) and the Permo-Triassic group (SAP_1, SAP_5, SAP_6, SAT_r).

Palaeozoic and Mesozoic geomagnetic poles available for South America; the polar wandering curve shown is that defined by Valencio and Vilas [1]; apparently the Jurassic-Cretaceous polar zone is independent of the Permo-Carboniferous 'age group' (SAC_1, $SAPC_1$, SAP_3, SAP_4) and the Permo-Triassic Group (SAP_1, SAP_5, SAP_6, SAT_r).

2.3. *The Mesozoic palaeomagnetic column*

McElhinny and Burek [10], have recently proposed a geomagnetic field polarity time scale for the Mesozoic; the authors have divided the Mesozoic into intervals of predominantly normal polarity and intervals of mixed polarity. The palaeomagnetic polarity and radiometric information is not sufficient to define with exactness the changes of polarity within the Cretaceous, and these authors indicate that the long normal interval during the Cretaceous, by them named Mercaton Magnetic Interval, extends from 'around' 120 my up to 70 my. The same authors have continued up into the Hauterivian, the reversed zones at the base of the Cretaceous, within their Mixed Polarity Serra Geral Zones (120–150 my), on the basis of information from Hungarian igneous rocks; however this information does not allow us to fix with exactness the boundary between the Mercaton Interval and the Mixed Interval where Serra Geral Zones are. Furthermore, the Mixed Interval of polarity named Serra Geral Zones (120–150 my) has been objected to by Creer [11]. This author introduces the Mesozoic palaeomagnetic column defined by USSR scientists, particularly by D.M. Pecherski, and concludes that the agreement between Pecherski's column and that of McElhinny and Burek "is rather good but not exact" (sic). Particularly, Pecherski's column for the Cretaceous seems to be more complete than that of McElhinny and Burek. However it must be noted that the ages of some of his reversed zones are fixed without the support of radiometric information. That is the particular case for the Chagilskaja and Shirabadskaja zones, fig. 6.

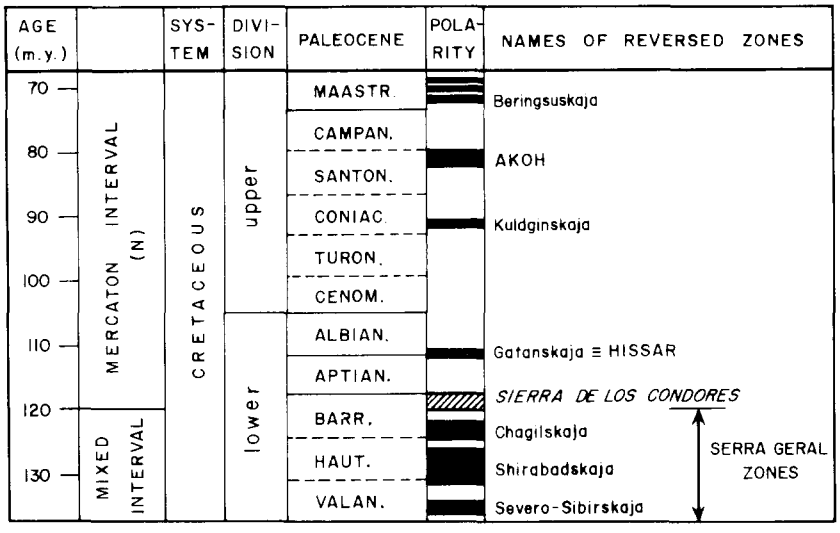


Fig. 6. Palaeomagnetic Polarity Column for the Cretaceous after McElhinny and Burek [10], Creer [11] and this paper. White regions indicate normal polarity, black regions reversed polarity. The reversed polarity zone defined in this paper (Sierra de los Cóndores) is shown by a striped region.

In this figure knowledge about the Cretaceous palaeomagnetic column in accordance with the last quoted papers is summarized; names given by McElhinny and Burek are given in vertical capital letters and those proposed by Creer in lower case letters. Both authors agree to suggest that a normal polarity period of the geomagnetic field is present between about 113 and 122 mybp. The Black-trachybasalt, lower flow, of the Vulcanitas Cerro Colorado Formation, has reversed thermoremanent magnetization (table 1, fig. 3d); as was mentioned before (subsect. 2.3) the mean potassium-argon age for this flow is 118.5 ± 6 my. Therefore the palaeomagnetic and radiometric studies of the Vulcanitas Cerro Colorado Formation of the Sierra de los Cóndores Group suggests the existence of a period of reversed polarity of the geomagnetic field about 119 my before the present time within a previously defined period of normal polarity. However, we must note that, as the accuracy of the dating is 6 my, there would be an overlapping between this period of reversed polarity and the next oldest reversed period (Chagilskaja, fig. 6). Then there are two possible interpretations for the palaeomagnetic and radiometric data of the Vulcanitas Cerro Colorado Formation: (i) they define the existence of a previously unrecognized period of reversed polarity of the geomagnetic field about 119 my before the present time, or (ii) the Chagilskaja reversed period continues up to the Aptian-Barremian boundary. As the available information is not sufficient to fix with exactness the boundaries of the Chagilskaja period (according to the author's knowledge there is no radiometric age supporting it), the author accepts that a new reversed period has been discovered at the lower Cretaceous; this reversed period has been indicated by stripes and named Sierra de los Cóndores in fig. 6. A new analysis of the evidence on the basis of which the age of Chagilskaja period was fixed would probably allow us to see whether the second interpretation is valid.

3. Interpretation

Cleaned remanent magnetizations of six time units are used to compute the palaeomagnetic pole (SAK_2) for the lower Cretaceous (118.5 ± 6 my) Vulcanitas Cerro Colorado Formation of the Sierra de los Cóndores Group. That suggests that SAK_2 is a geomagnetic pole; however we must consider the possibility that non-dipole components of the geomagnetic field be present affecting the position of SAK_2. This position, however is reasonably close to the Serra Geral Formation pole (SAK_1, fig. 5), and both poles define

with more precision the Lower Cretaceous pole position relative to South America: the more geomagnetic poles there are the better will be the precision of the mean polar position.

The mean distance between the sampling area of the Vulcanitas Cerro Colorado Formation (118.5 ± 6 my) and that of the Serra-Geral Formation (115–125 my) is about 1500 km. The difference between the palaeomagnetic poles for these two formations is not significant at the level $p = 0.05$ (fig. 4); this suggests that on average the geomagnetic field was similar to that of a geocentric dipole in Lower Cretaceous times and supports the interpretation that South America and Africa were separated in Lower Cretaceous time.

The Black-trachybasalt, lower flow, of the Vulcanitas Cerro Colorado Formation, radiometrically dated at 118.5 ± 6 my, have stable remanent magnetization with reversed polarity and defines the existence of a previously unrecognized period of reversed polarity within the Lower Cretaceous, here named Sierra de los Cóndores.

Acknowledgements

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